

Performance Testing

Date	19 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter
Type of Testing	Load Testing
Target Module / API	/predict endpoint (Flask prediction API)
Test Environment	Local (Flask development server)
Test Date	04 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits prediction form	1	10	Prediction returned successfully
2	10 concurrent users submit predictions	10	30	All requests handled without errors
3	50 concurrent users (load test)	50	60	Avg response stays under 2 sec
4	Sustained load test	25	300	System remains stable, no crashes

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 sec	Pass	Within target
2	Response Time (Max)	< 5 seconds	3.8 sec	Pass	Within target
3	Throughput (Req/sec)	> 50 req/sec	62 req/sec	Pass	Exceeds target
4	Error Rate	< 1%	0.3%	Pass	Well within limit
5	CPU Utilization	< 80%	68%	Pass	Acceptable load
6	Memory Utilization	< 80%	71%	Pass	Acceptable load

Step 4: Observations & Analysis

Key Findings:

The prediction endpoint responded quickly under normal and moderate load, with average response time staying well within the 2-second target even as concurrent users increased.

Bottlenecks Identified:

A slight increase in response time was observed above 50 concurrent users, likely due to the single-threaded Flask development server used for local testing.

Optimization Steps Taken:

Removed redundant feature computation in the prediction route and cached the trained model and scaler objects at app startup instead of reloading them on every request.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

